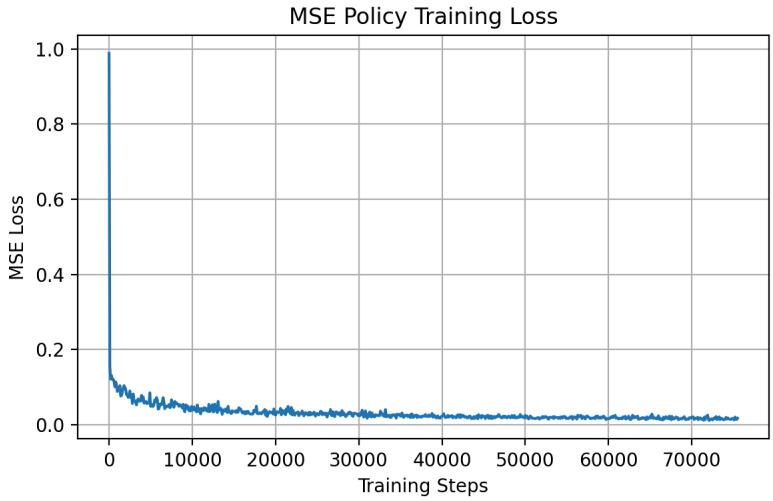
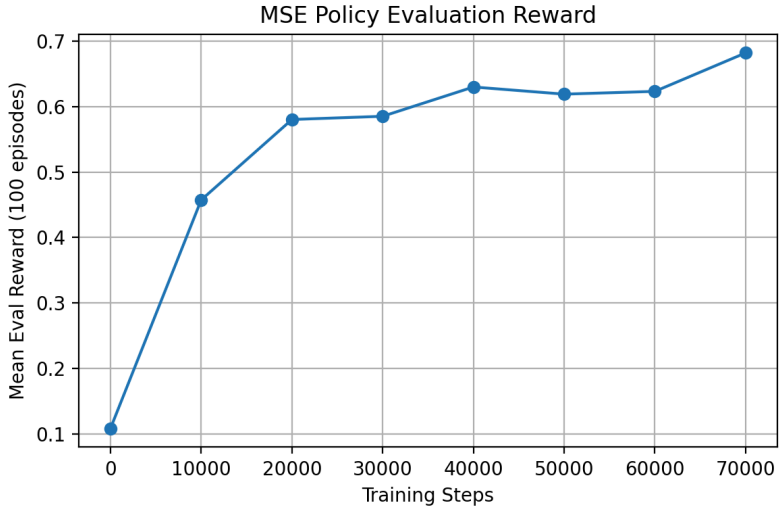
Homework 1 Report

MSE Policy:

Training steps vs. loss:



Training steps vs. rewards:

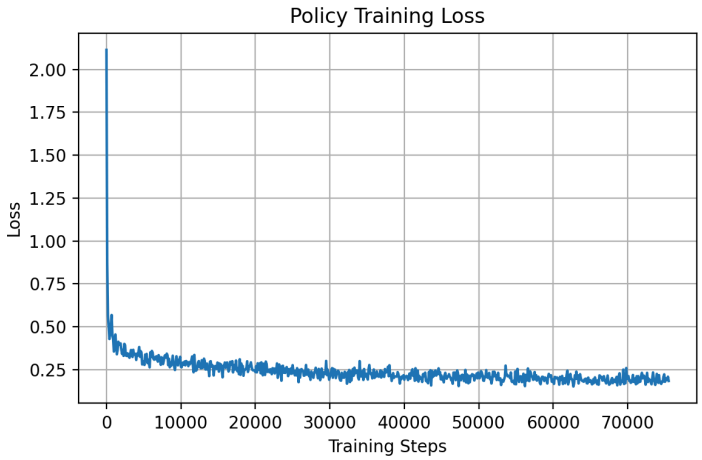


A brief description of your MLP architecture (number of layers, hidden size, activation functions, etc.)

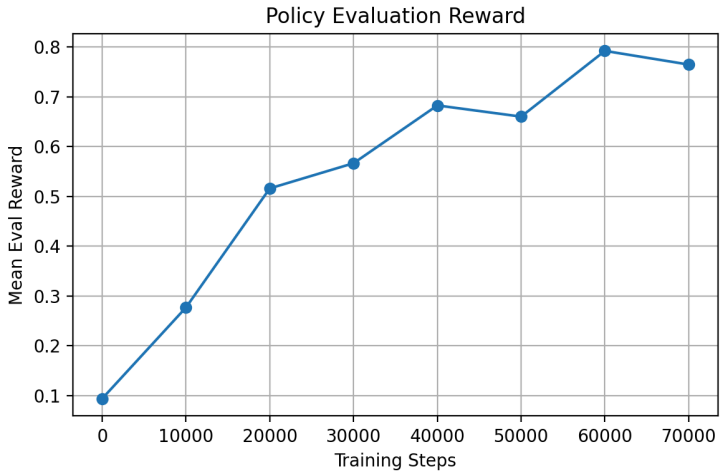
The MSE policy is implemented as a multilayer perceptron (MLP) that directly predicts a chunk of future actions from the current state. My architecture is fully connected MLP with 3 hidden layers(256 units per layer). It uses ReLU after each hidden layer and it has an output linear layer(no activation). The loss function is a Mean Squared Error (MSE) between predicted and expert action chunks.

Flow Matching:

Training steps vs. loss:



Training steps vs. rewards:



A brief qualitative description of how the flow matching policy behaves compared to the MSE policy.

The rollout videos evaluate every 10,000 steps. At step 0, the flow matching policy creates a larger range of jitter and wide exploration across the workspace. The cursor frequently changes direction and orientation between frames. This behavior occurs because action chunks are generated by integrating forward from pure Gaussian noise while the learned vector field is still untrained, resulting in random velocity updates. In contrast, the MSE policy produces smaller, more localized motions that remain closer to the dataset mean, leading to less spatial jitter.

By 40,000 steps, the flow matching policy shows significantly improved directional consistency, but small jitters are still visible, particularly when the cursor approaches the object. The MSE policy, while more stable in early motion, often commits to an imperfect action chunk and continues executing it open-loop, causing the cursor to stall or fail to recover once misaligned. At this stage, the flow matching policy achieves higher evaluation reward though.

At 70,000 steps, both policies successfully complete the task in most rollouts. However, from my observation, there's no major differences between the policies. Flow matching policy becomes smoother but still not as smooth as MSE policy. Also, MSE policy exhibits minor corrective motions at chunk boundaries, and also still got stuck in one of the rollouts. In successful rollouts, MSE will perform it much faster than flow matching policies. Overall, the flow matching policy is still more robust.